HP_power_board

File: HP_power_board.kicad_sch

LP_logic_board

File: LP_logic_board.kicad_sch

pi_power_stub

File: pi_power_stub.kicad_sch

pack_sense_board

File: pack_sense_board.kicad_sch

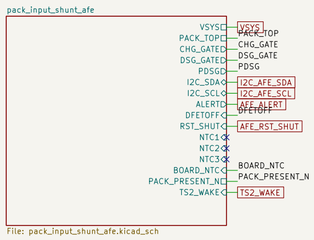
marengo

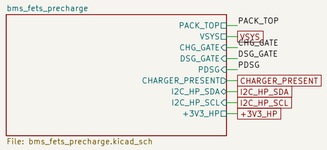
Sheet: /

File: marengo_pdu.kicad_sch

Title: marengo power board rev A

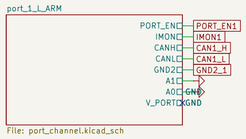
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 1/27

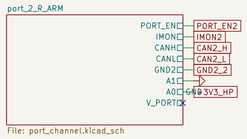


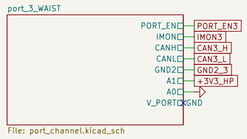


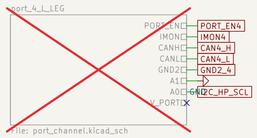


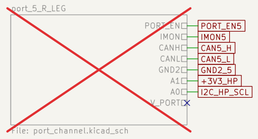


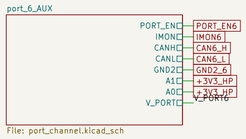




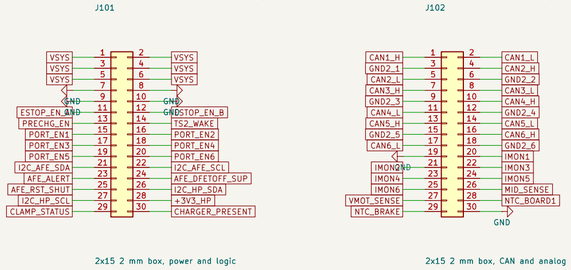


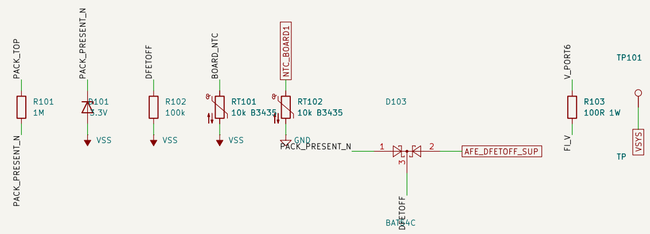






Board-to-board: two 2x15 2 mm ribbons to LP, pin for pin. No stock 2x30 latched header exists. All 60 pins used.





DFETOFF, active high (config 0x02): pack sense harness pulled, or supervisor backup, opens DSG. E-stop loop deliberately not wired here: it would deadlock boot (12 V comes after DSG) and drop compute.
Pull-up from PACK_TOP, not LP: it holds with the LP rails down. 54 uA through the presence loop when mated.
Fault injection: test mode, disarmed. Short H-L, H-GND2, L-GND2, H-VBUS via 100R.

marengo
Sheet: /HP_power_board/
File: HP_power_board.kicad_sch

Title: HP_power_board: high-power board

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 2/27

BAT from the pack + terminal, so the AFE stays up if J2 is pulled
with pack power still connected. Fig 16-2: diode, 100R, 1uF, 100pF. CP1 is 1uF, not 0.47uF: see sheet bms_fets_precharge. Caps 100 V.

PACK_TOP
D1
100V, verify
R15
100R
BAT
C14
1uF
VSS
C15
100pF
VSS
C16
1uF
CP1

J2
2x10 locking header
NTC1
NTC2
BAT-
BAT-
PACK_PRESENT_N
B2
B4
B6
B8
B10
B12
B1
B3
B5
B7
B9
B11
B13

J2: harness to the pack sense board 2x10, same pin map.
Columns 4-10 zigzag B0..B13: a bent pin shorts 1 cell
across a column, 2 cells along a row.
NTC return = B0 at the pack sense board. Spare: pin 12.
B0 lands on BAT- (datasheet p70: VC0 tied to VSS on PCB).

J1 pack +, J3 pack -: 10 AWG ring-lug pigtails to the XT90 at the pack.

J1
PACK_TOP
M4 terminal, REDCUBE 74651194R
J3
BAT-
M4 terminal, REDCUBE 74651194R
NT1
Net-Tie
VSS
SRP
SRN
RS1
0.3mR CSS4J-4026 5W
GND

Pack negative -> shunt -> system GND. SRP is the cell side;
charge current makes SRP positive (datasheet p5).
AFE VSS ties at BAT-, so VSS sits above GND by I x 0.3 mR
in discharge: 60 mV at the 200 A SCD point.

R1 20R B13 VC16 C1 220nF
R2 100R B12 VC12 C2 220nF
R3 100R B11 VC11 C3 220nF
R4 100R B10 VC10 C4 220nF
R5 100R B9 VC9 C5 220nF
R6 100R B8 VC8 C6 220nF
R7 100R B7 VC7 C7 220nF
R8 100R B6 VC6 C8 220nF
R9 100R B5 VC5 C9 220nF
R10 100R B4 VC4 C10 220nF
R11 100R B3 VC3 C11 220nF
R12 100R B2 VC2 C12 220nF
R13 100R B1 VC1 C13 220nF
R14 100R BAT- VC0

Cell filter: 100R + 220nF, 20R on VC16 (SLUAA81 2.5, top cell IR drop)
(datasheet p9: RC 20-100R, CC 0.1/0.22/1 uF). 100R gives the balance FET its gate drive.

U1
BQ76952PFBR
BAT 47
CP1 46
VC16 48
VC15 1
VC14 2
VC13 3
VC12 4
VC11 5
VC10 6
VC9 7
VC8 8
VC7 9
VC6 10
VC5 11
VC4 12
VC3 13
VC2 14
VC1 15
VC0 16
SRP_F 18
SRN_F 20
VSS 17
NC 19
NC 44
CHG 45 CHG_GATE
DSG 43 DSG_GATE
PCHG 40
PDSG 39 PDSG
PACK 42 PACK_S
LD 41 LD_S
FUSE 38
BREG 37
REGIN 36
REG1 35 VSS
REG2 34
REG18 24 C19 2.2uF VSS
SDA 27 I2C_SDA
SCL 26 I2C_SCL
ALERT 25 ALERT
RST_SHUT 33 RST_SHUT
DFETOFF 30 DFETOFF
CFETOFF 29 NTC3
HDQ 28 BOARD_NTC
DCHG 31
DDSG 32
TS1 21 NTC1
TS2 22 TS2_F
TS3 23 NTC2

13S per Table 16-3 (p75): cells 1, 2, 16 must be real.
VC0..VC12 = cells 1-12. VC15-VC16 = cell 13.
VC13-15 short to VC12. Cell inputs in use: 1-12 and 16.

PACK and LD both sense VSYS, the common port after the FETs
(Fig 16-3 R42-R45). PCHG, FUSE, DCHG, DDSG unused, left floating (p75).

PACK_S R18 5.1k R19 5.1k VSYS
LD_S R20 5.1k R21 5.1k VSYS
RST_SHUT R22 1k VSS
TS2_F R23 10k TS2_WAKE
NTC1 C20 1nF VSS
NTC2 C21 1nF VSS
NTC3 C22 1nF VSS

Four thermistors: TS1, TS3, CFETOFF (NTC1-3), HDQ (board).
TS2 stays free as SHUTDOWN wake (p22 pin table, section 13):
switch to VSS through 10k (Fig 16-2 R27); 1nF; p65 limit <4nF.
REGIN, BREG tied to VSS, REG1, REG2 open (p75); I2C pull-ups on LP.
CFETOFF as NTC needs DFETOFF set as BOTHOFF to stop CHG by pin (p48).

SRP R16 100R SRP_F
C17 100nF
C18 100pF
SRN R17 100R SRN_F

p65, p76: 100R each side, 100nF + 100pF differential, near U1,
symmetric, same PCB side. Kelvin SRP/SRN routed in parallel from RS1.

SIZING (shunt 0.3 mOhm: no 0.25 mOhm 4-terminal part exists: datasheet p22 steps)
OCD1 20 mV = 66.7 A (hardware comparator, up to 425 ms) OCD3 60 A for 1 s (current-based, inside the AFE, the agreed 60 A ceiling)
OCC 14 mV = 46.7 A at 425 ms SCD 60 mV = 200 A, +/-15% = 170-230 A, delay 15-450 us in 15 us steps (keeps the 150 A, 100 ms path rating)
OCD and OCC are set in 2 mV steps. Accuracy below 20 mV is -2/+2.65 mV (p22), about -6.7/+8.8 A on this shunt.

OPEN ON THIS SHEET
1. Balance FETs are captured below (SLUAA81 Fig 2-2). OTP/data memory owed: Cell Balance Min Cell V 3900 mV, Max Cells 4.
2. CP1 raised to 1 uF for the 4+4 FET gate load (p65 rule): worst case 1.69 uC droops 11 V by 1.7 V, pump start about 130 ms.
3. OTP owed: DFETOFF active high (0x02), CFETOFF and HDQ as thermistor inputs. Presence and DFETOFF logic are on HP_power_board.
4. RS1 land: CSS4J-4026 on the BVR4026 library footprint, check against the Bourns drawing.

External balance, SLUAA81A Fig 2-2: internal FET current through VC(n-1) input resistor gives Vgs. 3.9 V: 1.59 V min (RCB 46 R), FET rated at 1.5 V.
Rbal 27R: 140-150 mA external at 3.9-4.2 V, plus 17-19 mA internal. Cell Balance Max Cells 4: about 2.5 W total.

Q1 R24 BG1 27R 1W 2512 BD1 B1 B0 R25 1k VC0 D2 BZT52C5V1 BAT- SSM3K56FS
Q2 R26 BG2 27R 1W 2512 BD2 B2 B1 R27 1k VC1 D3 BZT52C5V1 B1 SSM3K56FS
Q3 R28 BG3 27R 1W 2512 BD3 B3 B2 R29 1k VC2 D4 BZT52C5V1 B2 SSM3K56FS
Q4 R30 BG4 27R 1W 2512 BD4 B4 B3 R31 1k VC3 D5 BZT52C5V1 B3 SSM3K56FS
Q5 R32 BG5 27R 1W 2512 BD5 B5 B4 R33 1k VC4 D6 BZT52C5V1 B4 SSM3K56FS
Q6 R34 BG6 27R 1W 2512 BD6 B6 B5 R35 1k VC5 D7 BZT52C5V1 B5 SSM3K56FS
Q7 R36 BG7 27R 1W 2512 BD7 B7 B6 R37 1k VC6 D8 BZT52C5V1 B6 SSM3K56FS
Q8 R38 BG8 27R 1W 2512 BD8 B8 B7 R39 1k VC7 D9 BZT52C5V1 B7 SSM3K56FS
Q9 R40 BG9 27R 1W 2512 BD9 B9 B8 R41 1k VC8 D10 BZT52C5V1 B8 SSM3K56FS
Q10 R42 BG10 27R 1W 2512 BD10 B10 B9 R43 1k VC9 D11 BZT52C5V1 B9 SSM3K56FS
Q11 R44 BG11 27R 1W 2512 BD11 B11 B10 R45 1k VC10 D12 BZT52C5V1 B10 SSM3K56FS
Q12 R46 BG12 27R 1W 2512 BD12 B12 B11 R47 1k VC11 D13 BZT52C5V1 B11 SSM3K56FS
Q13 R48 BG13 27R 1W 2512 BD13 B13 B12 R49 1k VC12 D14 BZT52C5V1 B12 SSM3K56FS

Pack terminals, sense header, shunt, BQ76952 AFE. Part numbers verify. TI SLUSE13B.

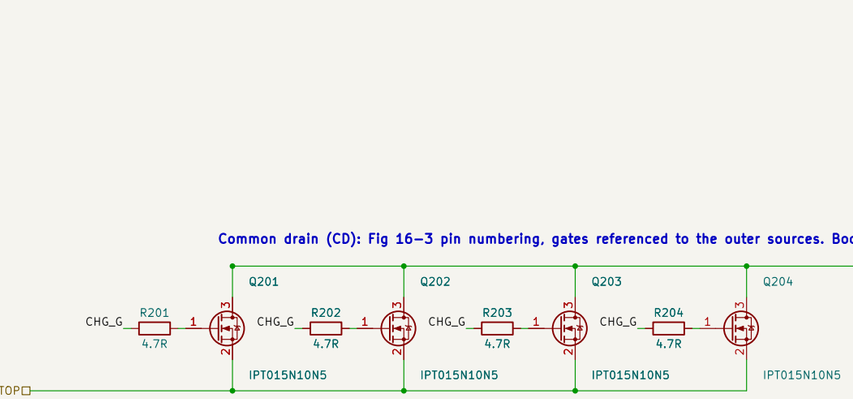
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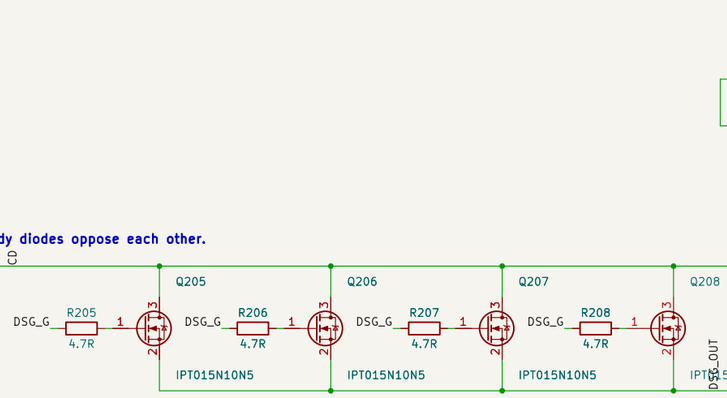
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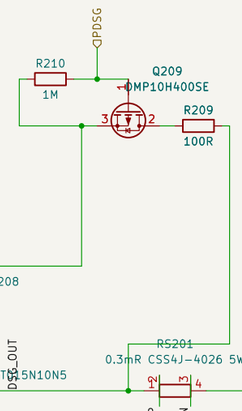
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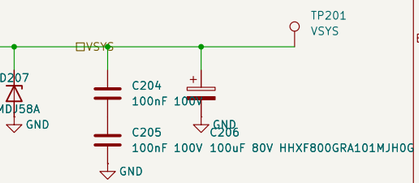
Size: A2 | Date: 2026-11-06 | Rev: A-wip
KiCad E.D.A. 10.0.8 | Id: 3/27

PDSG FET bypasses DSG only (Fig 16-1): source at CD. Driver sinks 30 uA, so gate-source >= 1M (p45).









CHG drive, Fig 16-3 R37/R31/D5.

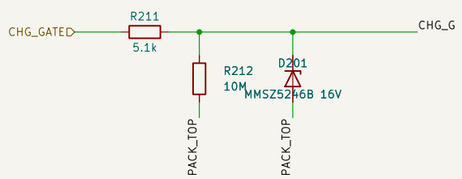
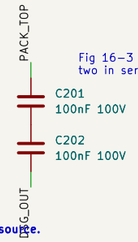
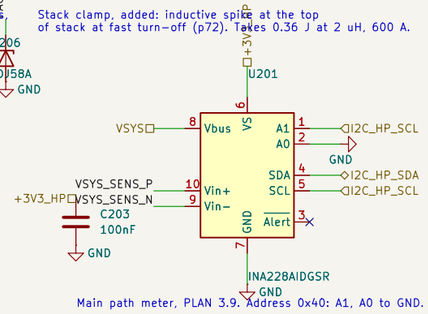


Fig 16-3 C28 C29: across the FETs, two in series.



Stack clamp, added: inductive spike at the top of stack at fast turn-off (p72). Takes 0.36 J at 2 uH, 600 A.

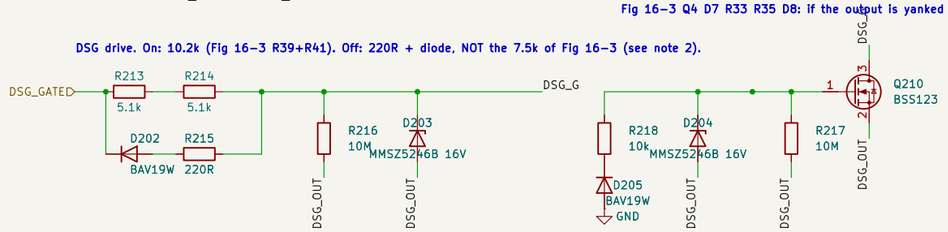




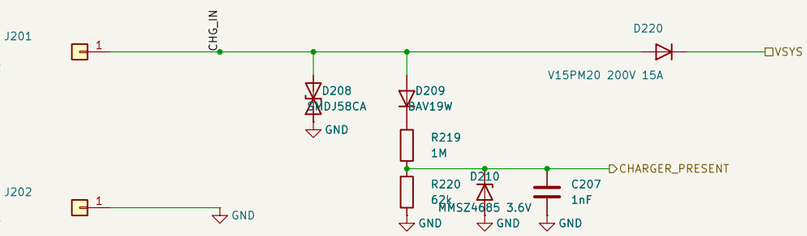
Main path meter, PLAN 3.9. Address 0x40: A1, A0 to GND. +/-40.96 mV range = +/-164 A.

Fig 16-3 Q4 D7 R33 R35 D8: If the output is yanked below GND, Q210 holds the DSG gate at its source.

DSG drive. On: 10.2k (Fig 16-3 R39+R41). Off: 220R + diode, NOT the 7.5k of Fig 16-3 (see note 2).



Charge input, common port: the charger joins on VSYS, DSG body diode side, as the BQ76952 expects (PACK pin sees it, p45). Also wakes the AFE from ship mode through LD.
PLAN 3.7 says reverse-polarity FET. With a reversed charger the blocker sees 54 + 54 = 108 V: above the 100 V FET and the LM74700 (75 V, SNOSD17G). So a 200 V Schottky: 0.7 V drop, 3 W at 4 A. See RATIONALE-log.
CHARGER_PRESENT: CHG_IN only, so VSYS cannot fake it: high above 34 V, 3.2 V at 54.6 V.



SIZING (design/README 2.2)
Conduction, 80 A: 2 x 0.6 mOhm hot x 80^2 = 7.7 W total. At 10 to 30 A: 0.1 to 1.1 W for these two banks.
FET: IPT015N10N5 Rds 1.5 mOhm max at Vgs 10 V, Vgs +/-20 V, so 11 V drive and 16 V zeners are in range (p69 needs >= 15 V).
Shunt voltage rule p69: 13S x 5 V = 65 V DC, 130 V peak. 100 V FETs meet DC. Peak: both TVS parts hold it under 100 V.

SOA must cover 600 A prospective for about 50 us, check at gate. Worked here, gate review to confirm:
1. Prospective current: 600 A / 4 = 150 A per FET. On-state part: 0.6 mOhm x 600 A = 0.36 V, 216 W bank, nothing.
2. Turn-off: TI's 4.3k path scaled to 4 x 12 nF has a Miller time of about 120 us at full bus: 100 us SOA is about 45 A per FET at 54 V. FAILS.
   With 220R: gate 48 nF, tau 10.6 us, plateau reached in 8 us, Miller 136 nC / 23 mA = 6 us (9 us max Qgd). 10 us SOA at 54 V is about 300 A per FET. PASSES at 150 A.
   Peak dT = 8.2 kW x 0.009 K/W (Zth 10 us) = 74 K. Margin shrinks with poor sharing at turn-off: scope it at bring-up.
3. Spike: p72 warns fast turn-off raises the top of stack. D206 clamps it. Tune 220R upward on the bench if the clamp runs hot.

CP1 is 1 uF on sheet 1: 8 x 211 nC max = 1.69 uC at p65's rule, droop 1.7 V from 11 V.
PDSG not PCHG: p45 PDSG charges the LOAD before DSG closes; PCHG charges a LOW BATTERY. README 2.1 described the PDSG job.
100R: peak 54.6^2/100 = 298 W. Stored energy 0.5 x 1.1 mF x 54.6^2 = 1.6 J per power-up, tau 0.11 s. Part must be pulse rated.
Set OTP: PREDISCHARGE on, timeout > 0.6 s (5 tau).

Fig 16-1, Fig 16-3 of TI SLUSE13B. High side, common drain, common port.

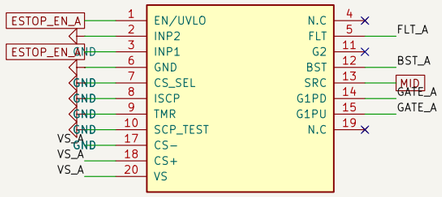
marengo

Sheet: /HP_power_board/bms_fets_precharge/
File: bms_fets_precharge.kicad_sch

Title: bms_fets_precharge: BMS CHG/DSG banks, predischarge, charge input

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 4/27

U301



TPS48100QDGXRQ1

R301 100R

C301 100nF 100V

C302 1uF 25V X7R

R302 100k

R303 100k

R304 1M

TP301

TP

FLT_A

VS_A

GND

GND

Bank A: VSYS to MID

Drain toward VSYS, source toward VMOT: body diodes conduct VMOT to VSYS, so regen still reaches the pack with the banks open (PLAN 3.4).

Q301

Q303

Q305

IPT015N10N5

IPT015N10N5

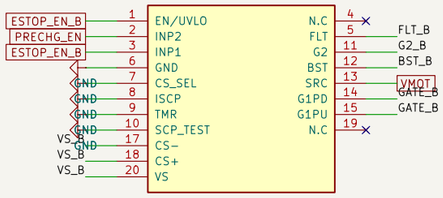
IPT015N10N5

R309 2.2R

R310 2.2R

R311 2.2R

U302



TPS48100QDGXRQ1

R305 100R

C303 100nF 100V

C304 1uF 25V X7R

R306 100k

R307 100k

R308 1M

TP302

TP

FLT_B

VS_B

GND

GND

Bank B: MID to VMOT

Q302

Q304

Q306

IPT015N10N5

IPT015N10N5

IPT015N10N5

R312 2.2R

R313 2.2R

R314 2.2R

VMOT precharge (new: IPT015N10N5 SOA is 0.7 A at 54 V, 10 ms, so a hard close on 1.8 mF is out). Close A, MID rises. PRECHG_EN high drives G2. Wait VMOT within 5 % of MID, then close B. EN low kills G2 too, so the loop gates precharge in hardware. Energy 2.68 J at 1.8 mF, tau 180 ms, peak 29.8 W.

Status (TPS48100 has no VDS output): MID, VMOT dividers 1M / 56k, 3.3 V = 62 V, plus VSYS on LP. Bank A short reads MID near VSYS with both open and VMOT low. FLT covers short circuit, charge pump UV, UVLO and test point, not routed to LP.

Q307

R315 100R PWR263S-3108-F

R316 100R PWR263S-3108-F

R317 1M

C305 100nF 100V

R318 1M

R319 56k

C306 1nF

R320 49.9R

R321 1M

R322 56k

C307 1nF

R323 49.9R

TP303

TP304

TP305

TP

TP

TP

GND

GND

GND

GND

GND

GND

TPS48100−Q1 simple−driver mode (SLVSGC0 Fig 7−11). No firmware in the enable path.

marengo

Sheet: /HP_power_board/estop_banks/

File: estop_banks.kicad_sch

Title: estop_banks: E−stop banks A and B, VMOT precharge

Size: A3	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 5/27

Bulk: 18 x 100 uF 80 V hybrid, 1.8 mF. 220 uF 80 V 10x12.5 does not exist. Can is 10 x 16.5 mm: footprint is the closest library match, real land owed.

C401 100uF 80V HHXF800GRA101MJH0G
C402 100uF 80V HHXF800GRA101MJH0G
C403 100uF 80V HHXF800GRA101MJH0G
C404 100uF 80V HHXF800GRA101MJH0G
C405 100uF 80V HHXF800GRA101MJH0G
C406 100uF 80V HHXF800GRA101MJH0G
C407 100uF 80V HHXF800GRA101MJH0G
C408 100uF 80V HHXF800GRA101MJH0G
C409 100uF 80V HHXF800GRA101MJH0G
C410 100uF 80V HHXF800GRA101MJH0G
C411 100uF 80V HHXF800GRA101MJH0G
C412 100uF 80V HHXF800GRA101MJH0G
C413 100uF 80V HHXF800GRA101MJH0G
C414 100uF 80V HHXF800GRA101MJH0G
C415 100uF 80V HHXF800GRA101MJH0G
C416 100uF 80V HHXF800GRA101MJH0G
C417 100uF 80V HHXF800GRA101MJH0G
C418 100uF 80V HHXF800GRA101MJH0G
D401 SMDJ58A
D402 SMDJ58A

C419 1uF 100V
C420 4.7uF 25V
R401 464k
R402 49.9k
R403 9.09k

U401
TPS7A4001DGNR

U403
LM4040AIM3-2.5

Clamp: 57.50 V on, 56.50 V off (VOH 11.7 V, 19.1M hysteresis). Spread 56.82–58.19 V full temperature, 56.99–58.02 V 0–60 C. 10 Ohm 50 W clad external: 330 W, 5.75 A.
FET, 2 mOhm shunt (INA228 SHUNT_CAL 2000, ADCRANGE 0), 30 A Schottky and M4 terminals are sized for the leg stage 2 Ohm resistor, 29 A, 1.7 kW short time: only the external resistor swaps.
INA228 0x4D: clamp power and energy on its own accumulator. Budget about 5 mA, 0.23 W in the LDO.
Failure: stuck-on 330 W caught by brake NTC and opening the E-stop banks. SMDJ58A stays off under 64.4 V, spike protection only.
Regen decision (coordinator 11-06): 10 Ohm in rev A. 30 A regen against 5.75 A clamp: BMS opening mid-brake rises 13.5 V/ms, drives stop at 60 V in about 185 us, torque drops. Leg stage fits 2 Ohm.

R404 499k 0.1%
R405 619k 0.1%
R406 51.1k 0.1%
R407 19.1M
C421 100pF C0G
C422 100nF

U402
TLV1805DBVR

Q402
BSS123

J401
J402
M4 terminal, brake R +
M4 terminal, brake R –
J403
brake R NTC

C423 1uF 25V
R408 4.7R
R409 100k

U404
UCC27517DBVR

Q401
IPT015N10N5

RS401
2mR CSS4J-4026K-2L00F 6W

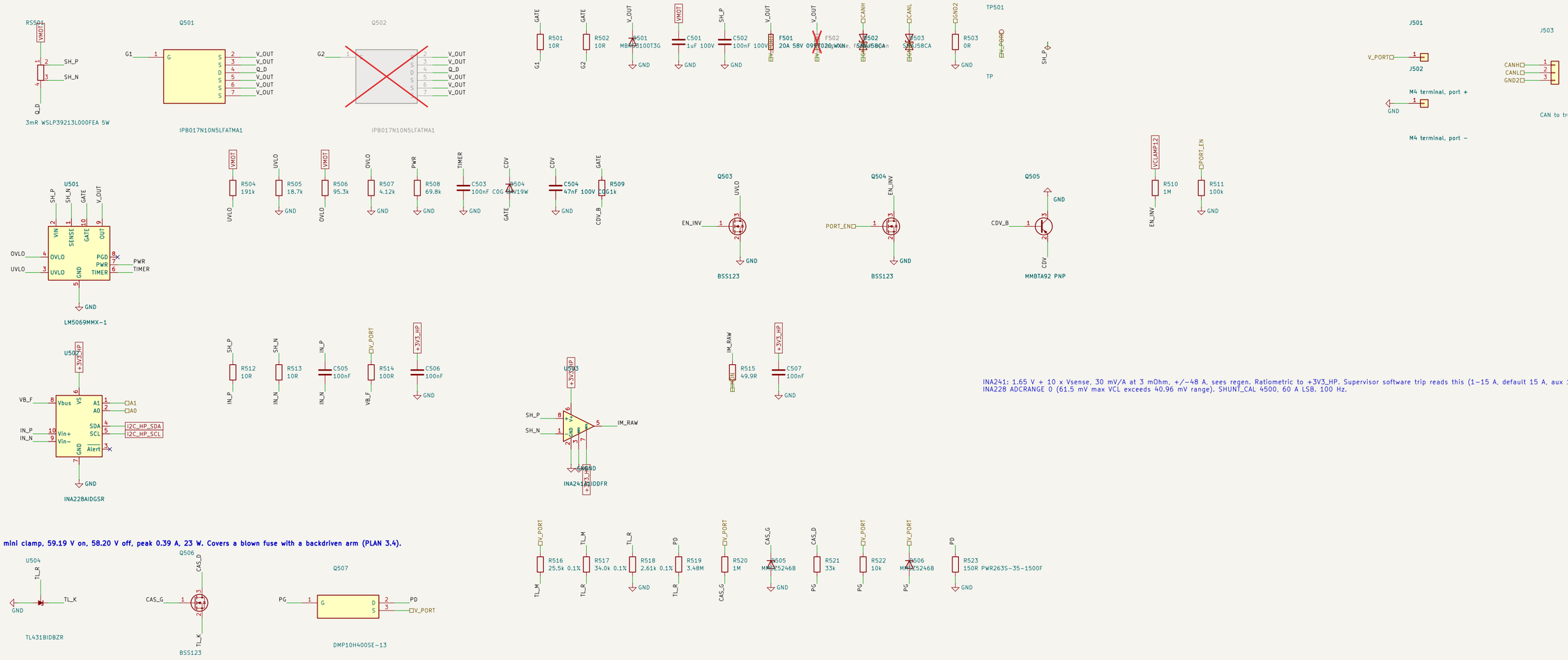
D403
MBRB30H100CT-E3/81

R410 10R
R411 10R
C424 100nF
R412 100R
C425 100nF

U405
INA228AIDGSR

Regen clamp powers itself from VMOT, works with supervisor, BMS and pack all off (PLAN 3.4).

marengo	
Sheet: /HP_power_board/vmot_bulk_clamp/	
File: vmot_bulk_clamp.kicad_sch	
Title: vmot_bulk_clamp: VMOT bulk, TVS, self-powered regen clamp	
Size: A2	Date: 2026-11-06
KiCad E.D.A. 10.0.8	Id: 6/27
	Rev: A-wip



Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3_100k down) turns Q3 on, Q2 off, UVLO released, Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

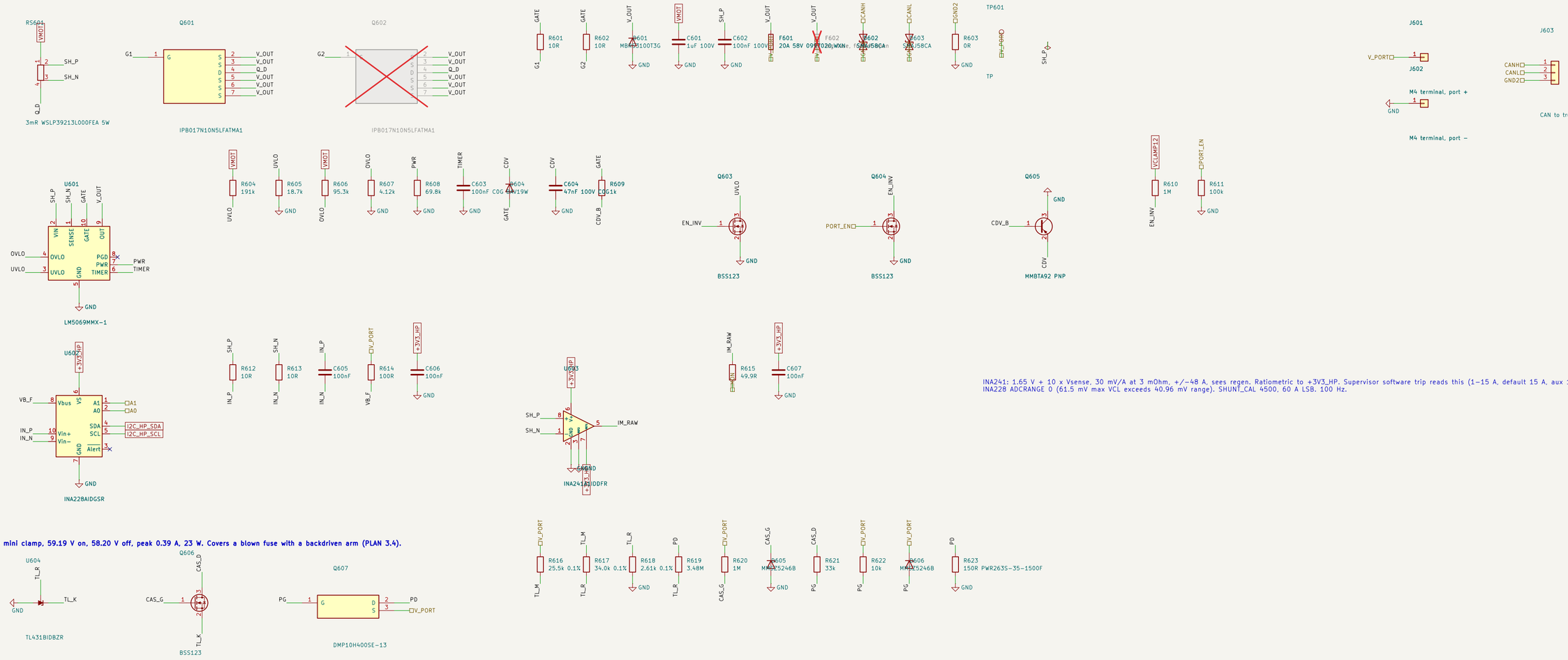
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.

marengo

Sheet: /HP_power_board/port_1_L_ARM/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 7/27



Load−side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

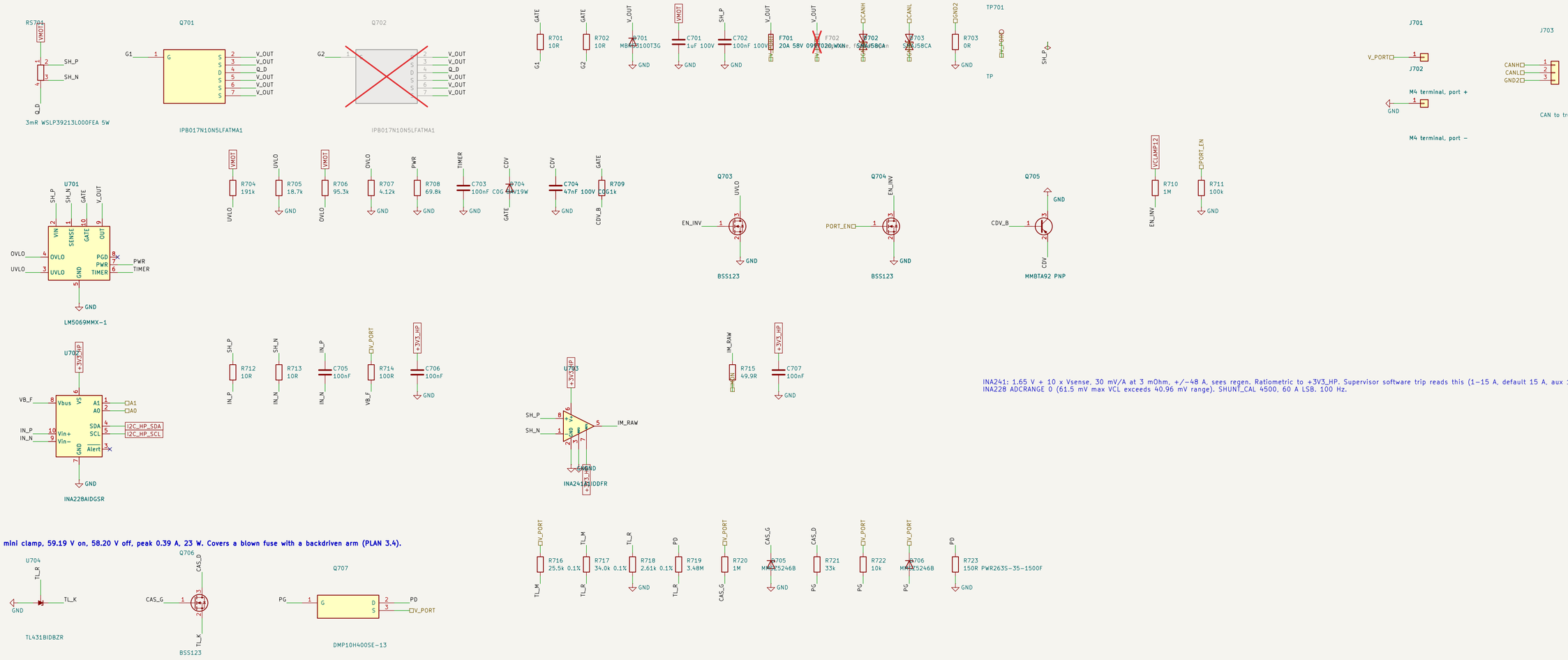
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.

marengo

Sheet: /HP_power_board/port_2_R_ARM/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 8/27



INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load-side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN-to-VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26–32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

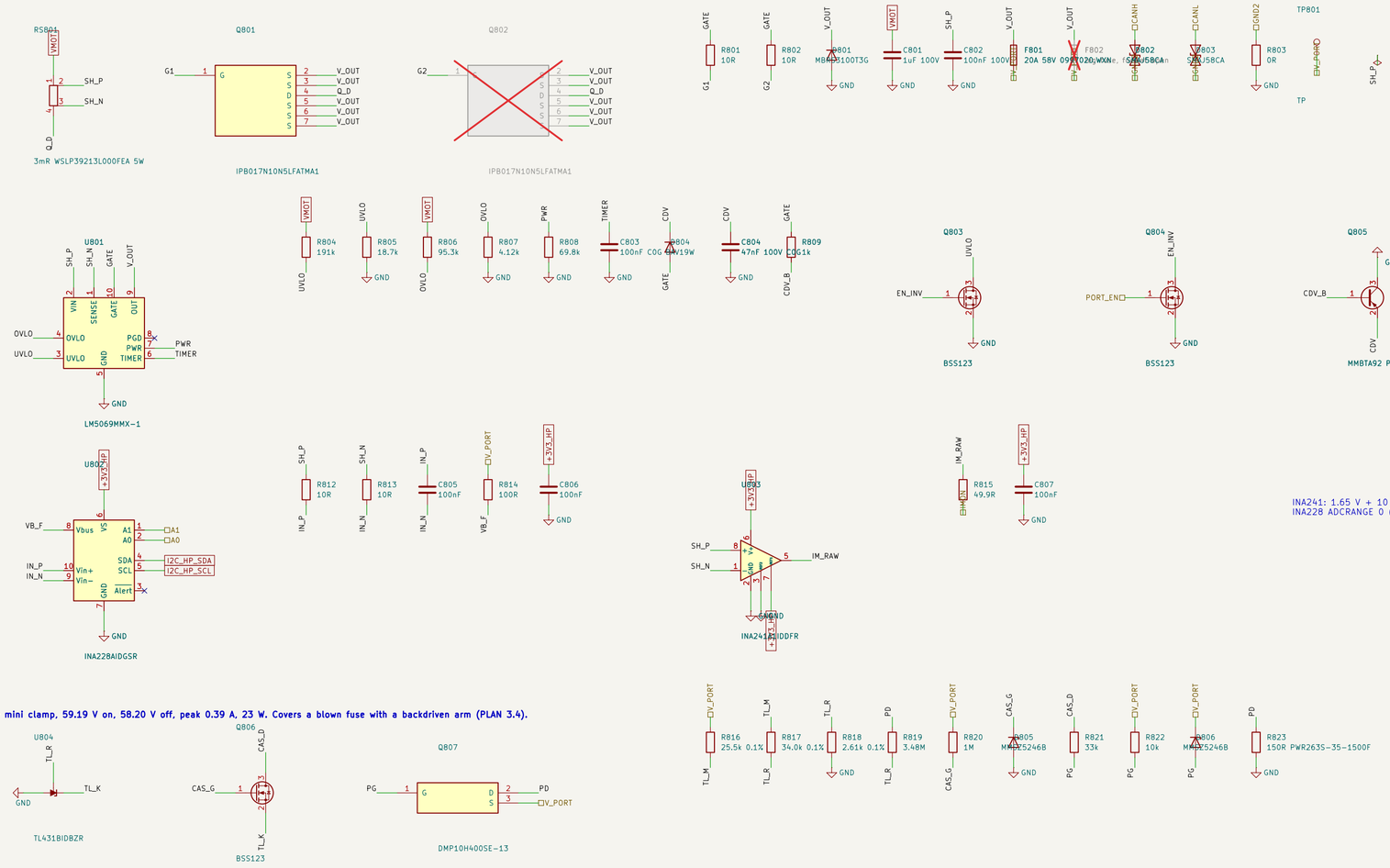
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.

marengo

Sheet: /HP_power_board/port_3_WAIST/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 9/27



Load−side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released, Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return: it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

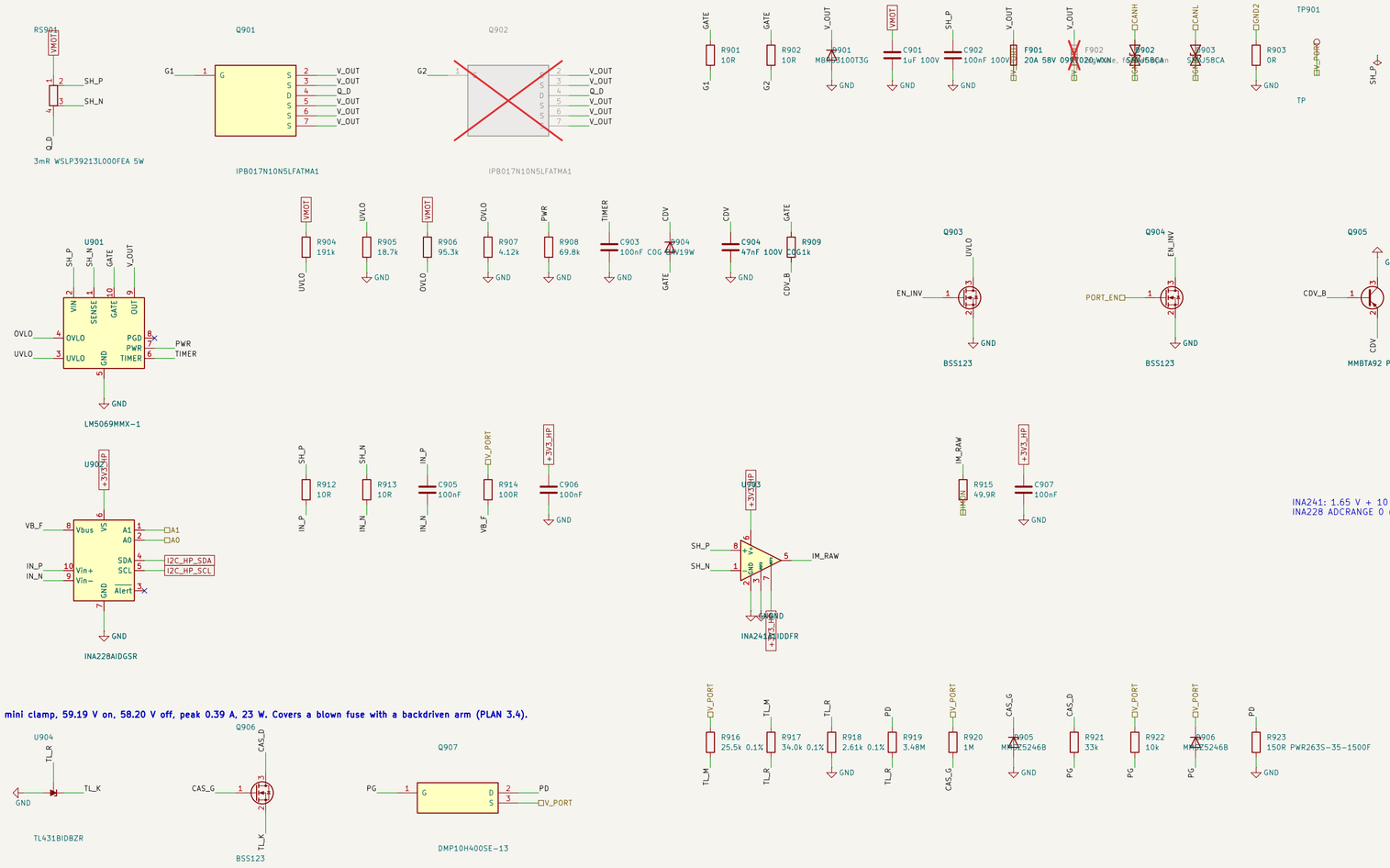
LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.

marengo

Sheet: /HP_power_board/port_4_L_LEG/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 10/27



INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1–15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load-side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2–20.5 A (48.5–61.5 mV / 3 mOhm), breaker 26.7–43.3 A, RPWR 69.8k = 200 W, fault timer 2.8–9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5–61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return; it does not block a CAN-to-VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26–32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

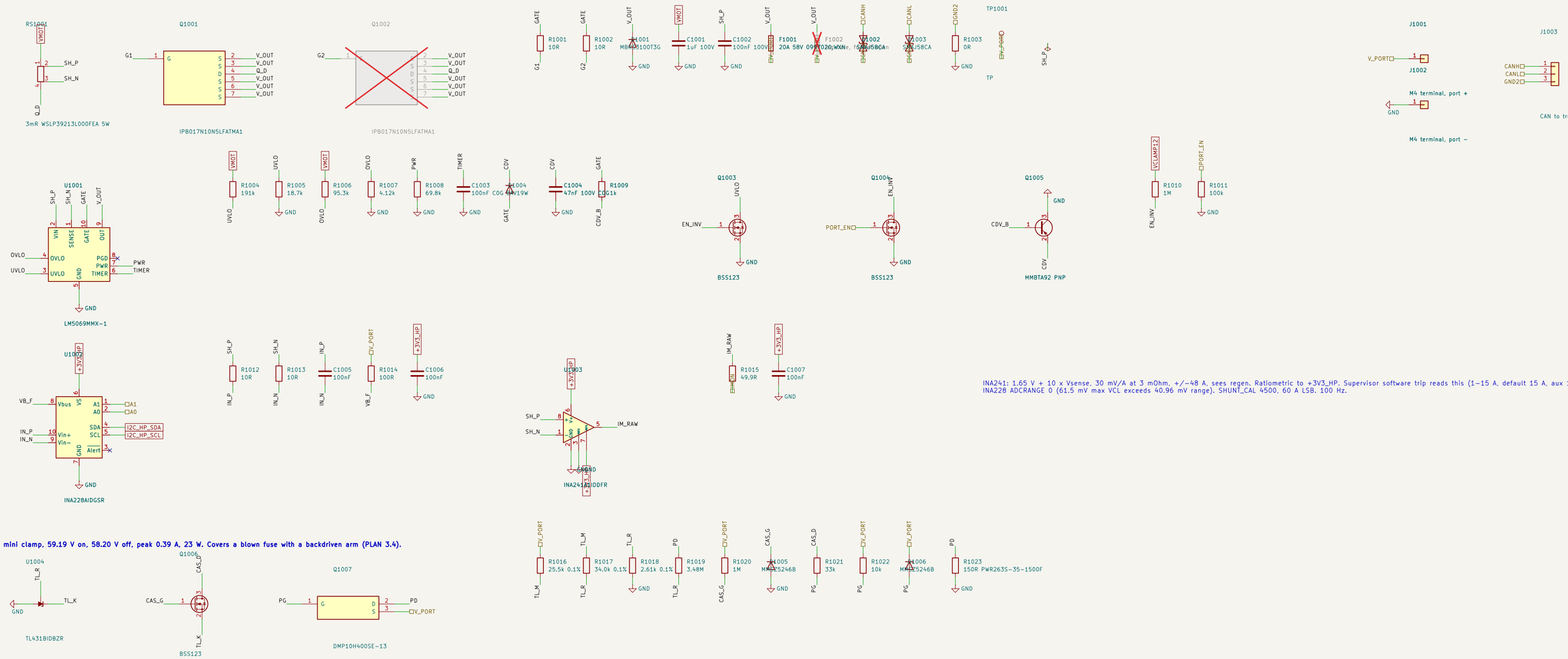
LM5069-1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load-side mini clamp.

marengo

Sheet: /HP_power_board/port_5_R_LEG/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 11/27



INA241: 1.65 V + 10 x Vsense, 30 mV/A at 3 mOhm, +/−48 A, sees regen. Ratiometric to +3V3_HP. Supervisor software trip reads this (1−15 A, default 15 A, aux 10 A), drops PORT_EN.
INA228 ADCRANGE 0 (61.5 mV max VCL exceeds 40.96 mV range). SHUNT_CAL 4500, 60 A LSB. 100 Hz.

Load−side mini clamp, 59.19 V on, 58.20 V off, peak 0.39 A, 23 W. Covers a blown fuse with a backdriven arm (PLAN 3.4).

Values shown: arm, waist, aux. LM5069 limit 16.2−20.5 A (48.5−61.5 mV / 3 mOhm), breaker 26.7−43.3 A, RPWR 69.8k = 200 W, fault timer 2.8−9.0 ms.
Real overload limit is the supervisor software trip at 15 A: XT30 and enclosed 16 AWG are treated as 15 A continuous. The 20 A fuse is only the backstop for a shorted FET.
Leg channels (DNP in rev A): Q502 fitted; RS501 1 mOhm (48.5−61.5 A); RPWR 23.7k; INA228 SHUNT_CAL rescaled. Leg fuse format open: DNP holder here beside F501. 997 MINI stops at 30 A.
Enable: PORT_EN high (3V3, 100k down) turns Q3 on, Q2 off, UVLO released. Supervisor dead: Q2 on, UVLO low, port off. Latch reset: drop PORT_EN, wait 148 ms, raise it.
UVLO on 32.0 V off 28.0 V, OVLO off 60.3 V. Starts on dV/dt: 47 nF covers 3.6 mF of drive capacitance.
GND2: bus reference ties to the port return here. Galvanic isolation then separates LP logic ground from motor return: it does not block a CAN−to−VBUS fault, ISOW1044 survives that at +/−58 V.
SMAJ58CA for sustained 54 V survival (NUP2105L VBR 26−32 V fails). Its clamp is above the ISOW1044 58 V limit and its capacitance is unchecked: scope on the bus.

LM5069−1 latched hot swap, INA228 + INA241 on one shunt, blade fuse, load−side mini clamp.

marengo

Sheet: /HP_power_board/port_6_AUX/
File: port_channel.kicad_sch

Title: port_channel: one limb port (instanced x6, legs DNP)

Size: A2	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 12/27

















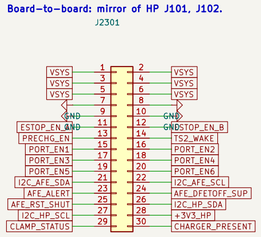


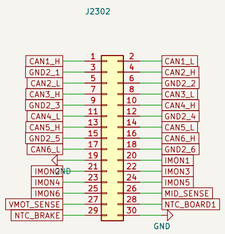






Board-to-board: mirror of HP J101, J102.



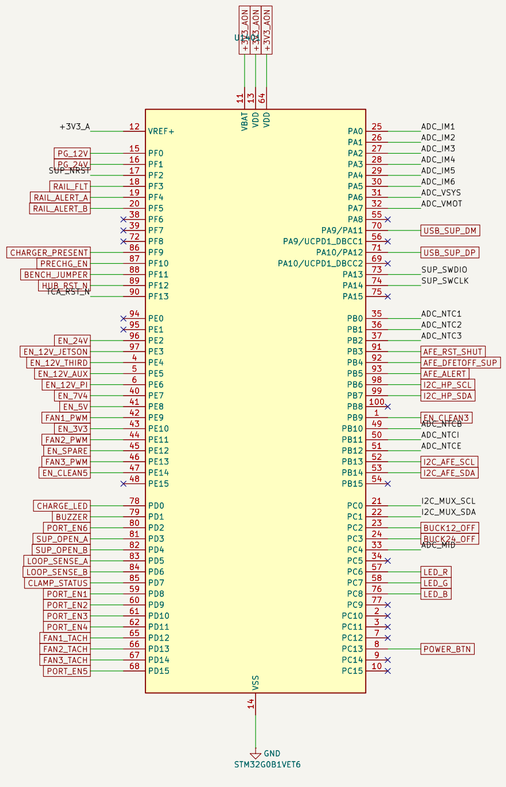


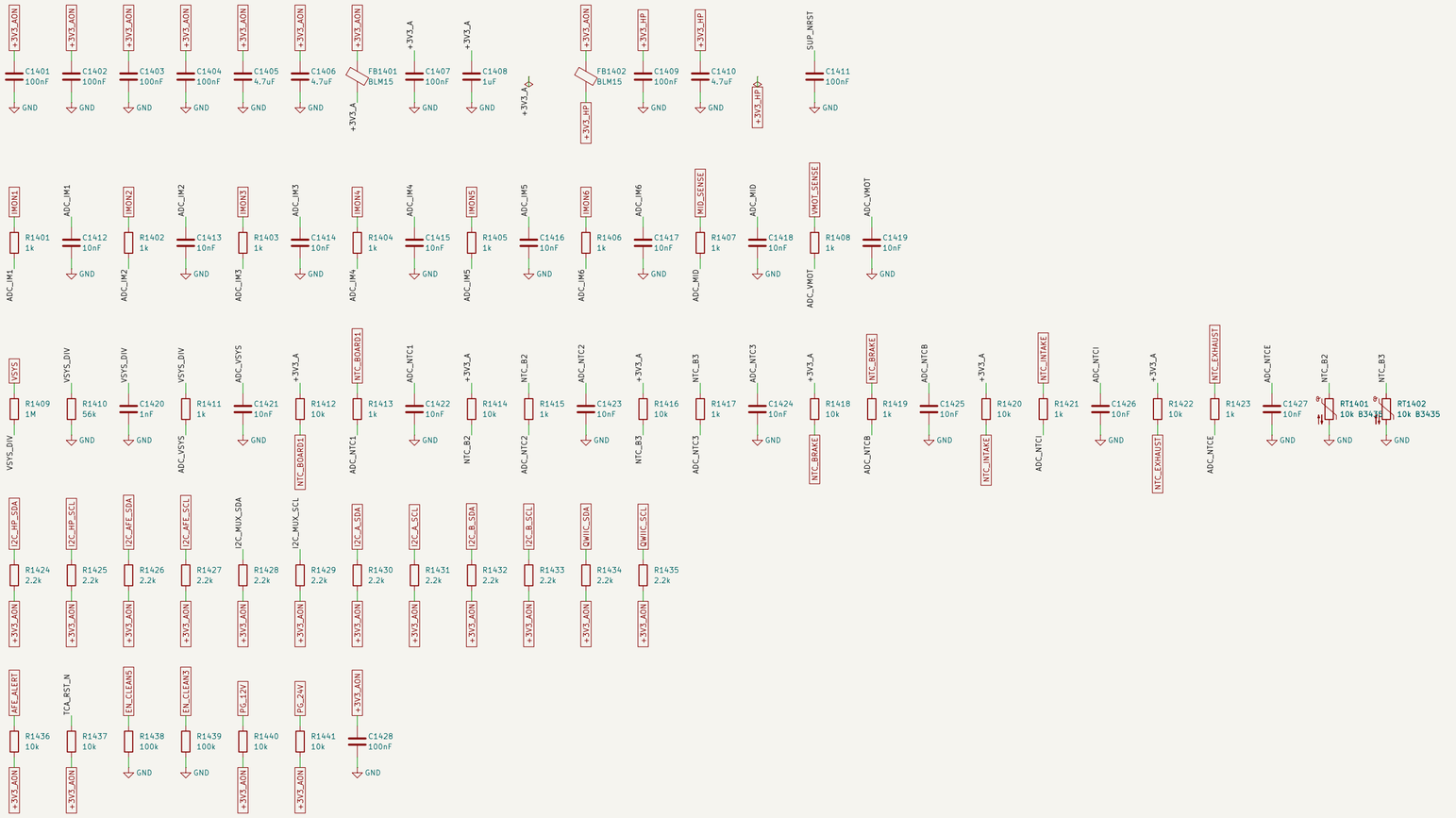
marengo

Sheet: /LP_logic_board/
File: LP_logic_board.kicad_sch

Title: LP_logic_board: low-power board

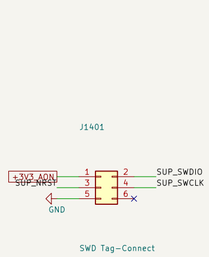
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 13/27

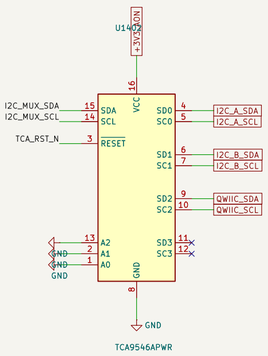




G0B1VET6 pin plan: parts check 68 of 68 against ST pin data. Extra lines beyond the plan: PC2/PC3 buck off, PF0/PF1 PG, PF3–5 rail flags, PB9 EN_CLEAN3. PF10 was BOARD_PRESENT, now PRECHG_EN. PC5 free.
I2C: 3 hardware buses. I2C1 HP meters. I2C2 AFE. I2C3 to TCA9546A at 0x70 (ch0 meters A, ch1 meters B, ch2 Qwiic, ch3 spare).
Five NTC channels sit on 10k to the ADC reference. NTC_BOARD1 also reads open when the HP board is missing: replaces BOARD_PRESENT.

12 V LOAD SHED (firmware; budget sums to 22 A on a 12.5 A converter, DC limit 14.2 A min). Estimate: sum of 12 V rail meters and POL meters. Start values, tune on the bench, 500 ms hold, 1 A hysteresis.
1. Above 11.0 A: fans to reduced duty (FANx_PWM).
2. Above 11.5 A: third board off (EN_12V_THIRD).
3. Above 12.0 A: cameras and spare off (EN_12V_AUX, EN_SPARE).
4. Above 13.0 A, or PG_12V falling: PI stub and Jetson last (EN_12V_PI, EN_12V_JETSON), after telling the host.
PORT SOFTWARE TRIP (firmware): each ADC_IMx, 1–15 A per port, default 15 A on arm and waist, 10 A on aux. Over the limit drops PORT_ENx, then reset needs the 148 ms latch sequence.





Supervisor can open the banks and ports, never close them against the loop. All enables fail off.

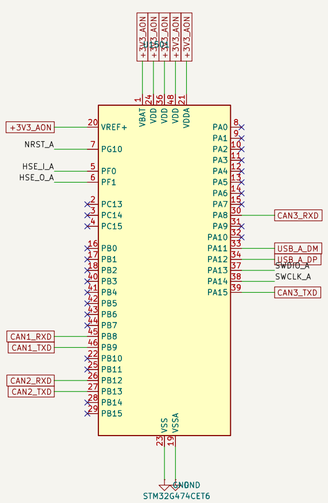
marengo

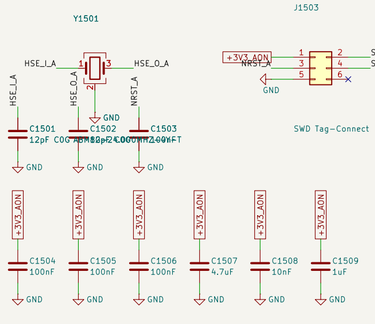
Sheet: /LP_logic_board/supervisor_mcu/
File: supervisor_mcu.kicad_sch

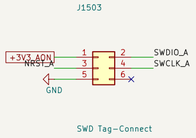
Title: supervisor_mcu: STM32G0B1 supervisor, I2C mux

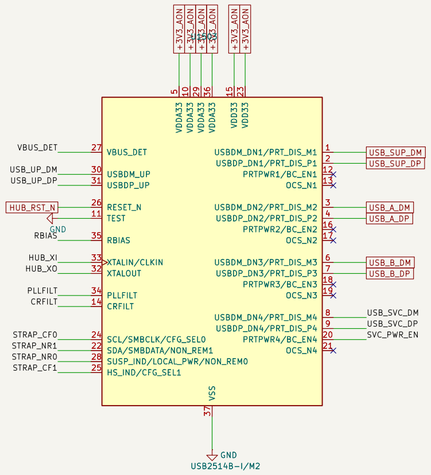
Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 14/27

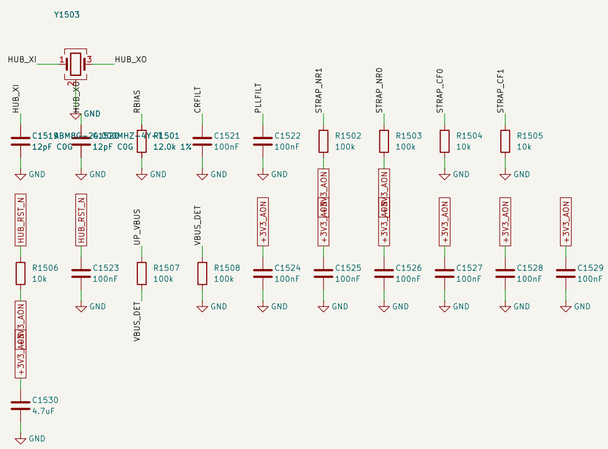
Hub: DN1 supervisor, DN2 bridge A, DN3 bridge B, DN4 service. No EEPROM: CFG_SEL 00, NON_REM 11, self powered, individual switching. TEST to GND. OCS_N1–4 open (internal pull-up).
FDCAN1_RX on PB8-BOOT0: program option bytes to ignore the pin before first run. PA15 released from JTDI for FDCAN3_TX. Firmware: budgetcan_fw, candleLight has no G474 build.
Service USB-C sits on a hub downstream port, so it is a host port: a laptop cannot reach the board through it. Rp 56k, default USB current.

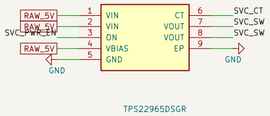


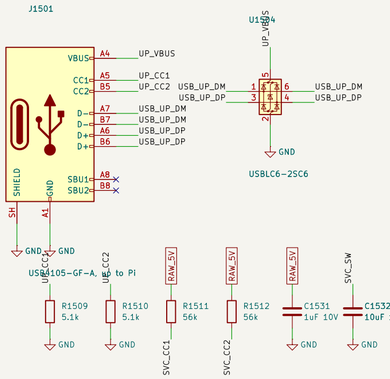


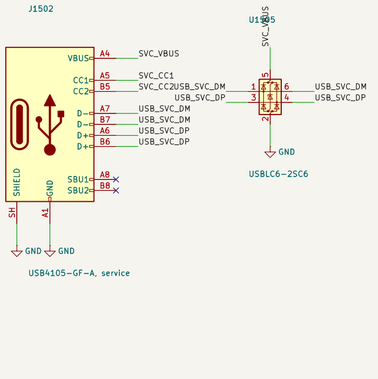












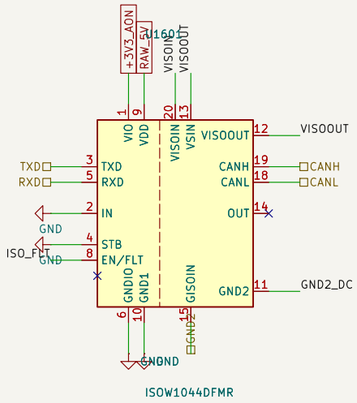
6 CAN buses over one USB-C to the Pi. Hub comes up without the supervisor, so the Pi keeps CAN if

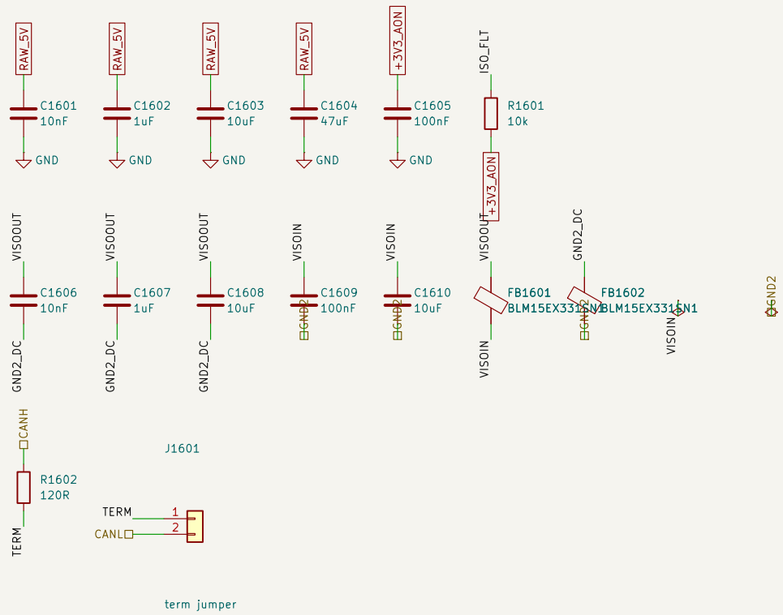
marengo

Sheet: /LP_logic_board/can_bridges_usb_hub/
File: can_bridges_usb_hub.kicad_sch

Title: can_bridges_usb_hub: two STM32G474 CAN bridges, USB2514B hub

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 15/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull−up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

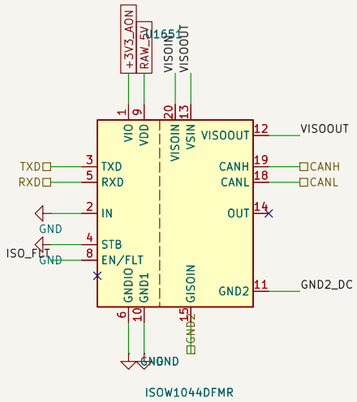
ISOW1044: integrated isolated supply. 4 mm metal keep−out around pins 11 and 12.

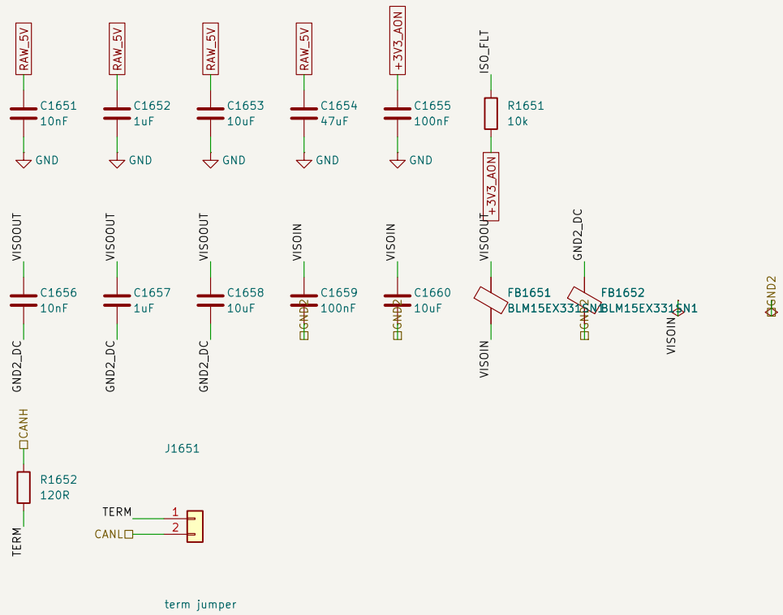
marengo

Sheet: /LP_logic_board/isolated_can_1/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 16/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

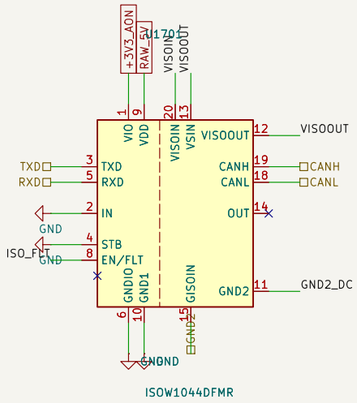
ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

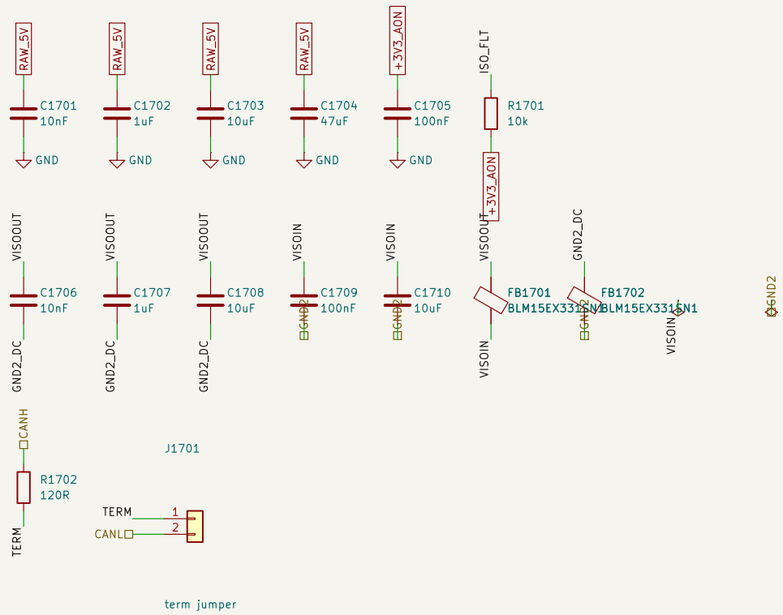
marengo

Sheet: /LP_logic_board/isolated_can_2/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 17/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

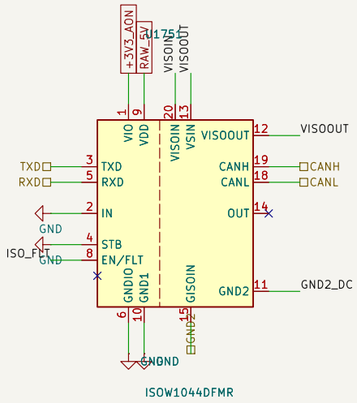
ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

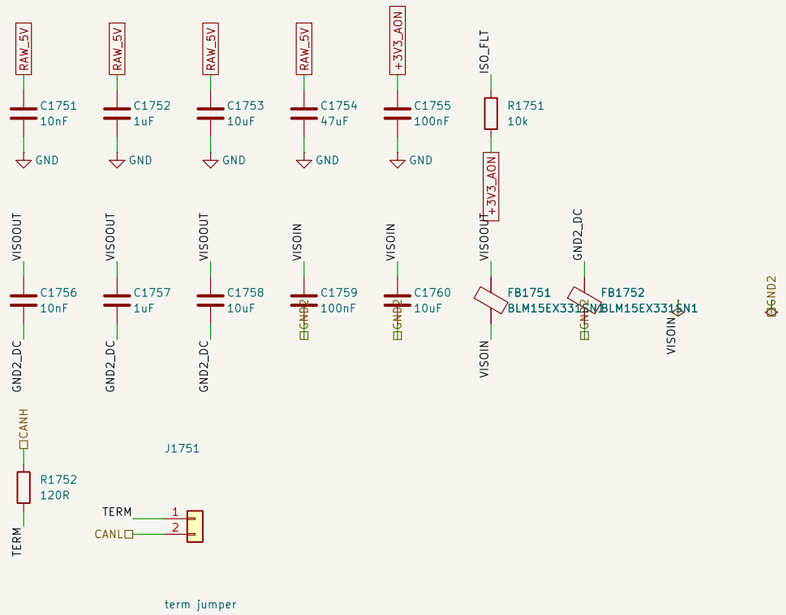
marengo

Sheet: /LP_logic_board/isolated_can_3/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 18/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull−up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

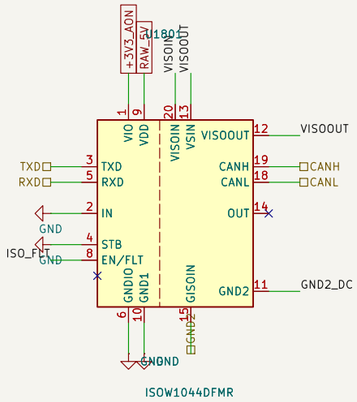
ISOW1044: integrated isolated supply. 4 mm metal keep−out around pins 11 and 12.

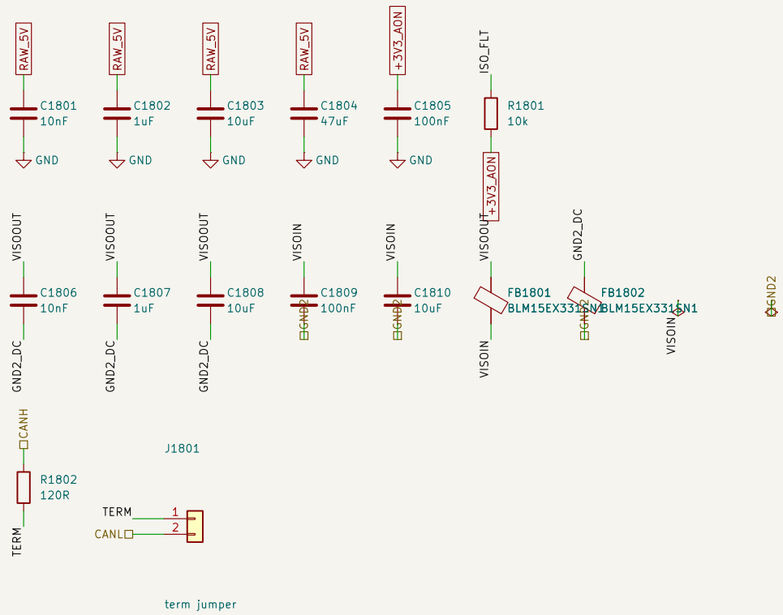
marengo

Sheet: /LP_logic_board/isolated_can_4/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026−11−06	**Rev: A−wip**
KiCad E.D.A. 10.0.8		Id: 19/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

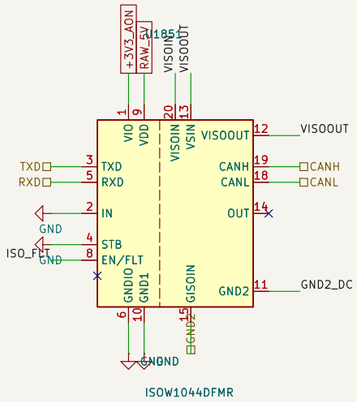
ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

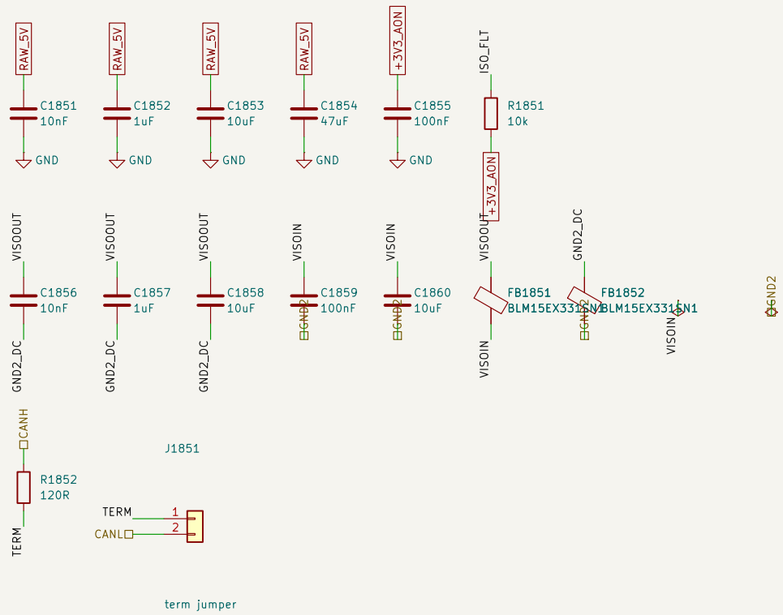
marengo

Sheet: /LP_logic_board/isolated_can_5/
File: isolated_can.kicad_sch

Title: isolated_can: one isolated CAN channel (instanced x6)

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 20/27





VDD 5 V is required: 6 ports draw up to 0.74 A from RAW_5V. Bus pins +/−58 V abs, +/−45 V differential. STB tied to GNDIO for normal mode. IN grounded, OUT open: spare isolated GPIO.
EN/FLT 10k pull-up: enabled, fault readable with a scope. Termination 120 Ohm by jumper per port, far end at the trunk.

ISOW1044: integrated isolated supply. 4 mm metal keep-out around pins 11 and 12.

marengo

Sheet: /LP_logic_board/isolated_can_6/		
File: isolated_can.kicad_sch		
Title: isolated_can: one isolated CAN channel (instanced x6)		
Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 21/27

Ideal diode and hold-up. LM5050-1 not LM74700: PLAN 2.3 wants 75 V; LM74700 stops at 65 V, and VSYS needs no reverse-polarity function here.
20 x 68 uF 80 V hybrid, 10 x 10.8 mm = 1.36 mF, fits the 13 mm board gap: 12.6 ms at 90 W from 46.8 V, 6.8 ms from 37 V. README claimed 1 mF for 10 ms: true only above 48 V.
The 24 V converter taps VSYS ahead of the diode so 75 W does not drain the hold-up.

U1101
LM5050MK-1
Q1101
CSD19532Q5B
C1101 100nF 100V
C1102 100nF 100V
D1101 B340 30V
Q1102 Q1103 CSD19534Q5A
U1102
Q1104
LM5145RGYR
BSS123
BUCK12_OFF
R1111 100k
L1101 XAL1510-682MED 33.2k
R1101 33.2k
R1102 33.2k
R1103 2.37k
R1104 6.04k
C1123 10nF
C1124 33pF
R1105 100R
C1125 1nF
C1126 33nF
C1127 100nF 25V
R1106 2.2R
C1128 2.2uF 25V
R1107 348R
C1129 18pF C0G 100V
R1108 1.21M
R1109 49.9k
C1130 100nF 100V
R1110 20k
C1131 4.7uF 100V
C1132 4.7uF 100V
C1133 4.7uF 100V
C1134 4.7uF 100V
C1135 4.7uF 100V
C1136 22uF 25V
C1137 22uF 25V
C1138 22uF 25V
C1139 22uF 25V
C1140 22uF 25V
C1141 22uF 25V
C1142 22uF 25V
U1103
Q1105 Q1106 CSD19534Q5A
Q1107
BUCK24_OFF
R1122 100k
L1102 XAL1510-333MED 33.2k
R1112 33.2k
R1113 80.6k
R1114 2.8k
R1115 17.8k
C1143 6.8nF
C1144 10pF
R1116 100R
C1145 680pF
C1146 33nF
C1147 100nF 25V
R1117 2.2R
C1148 2.2uF 25V
R1118 301R
C1149 22pF C0G 100V
R1119 499k
R1120 18.2k
C1150 100nF 100V
R1121 20k
C1151 4.7uF 100V
C1152 4.7uF 100V
C1153 4.7uF 100V
C1154 10uF 50V
C1155 10uF 50V
C1156 10uF 50V
C1157 10uF 50V
C1158 10uF 50V
C1159 10uF 50V
C1160 10uF 50V
C1103 68uF 80V B40950B0686M100
C1104 68uF 80V B40950B0686M100
C1105 68uF 80V B40950B0686M100
C1106 68uF 80V B40950B0686M100
C1107 68uF 80V B40950B0686M100
C1108 68uF 80V B40950B0686M100
C1109 68uF 80V B40950B0686M100
C1110 68uF 80V B40950B0686M100
C1111 68uF 80V B40950B0686M100
C1112 68uF 80V B40950B0686M100
C1113 68uF 80V B40950B0686M100
C1114 68uF 80V B40950B0686M100
C1115 68uF 80V B40950B0686M100
C1116 68uF 80V B40950B0686M100
C1117 68uF 80V B40950B0686M100
C1118 68uF 80V B40950B0686M100
C1119 68uF 80V B40950B0686M100
C1120 68uF 80V B40950B0686M100
C1121 68uF 80V B40950B0686M100
C1122 68uF 80V B40950B0686M100
Hold-up bank, 20 x 68 uF, VSYS_OR to GND

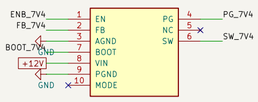
BUCK12_OFF, BUCK24_OFF: pulled up by the always-on rail so a sleeping or reset supervisor holds both converters off. Supervisor drives low to start them (Standby state, PLAN 3.2).
The 12 V converter is the 300 kHz master; SYNCOUT drives the 24 V converter 180 degrees out of phase. 12 V feeds its own VCC through a Schottky after start-up (lower loss).

3.3 V always-on. EN tied to VIN: on whenever VSYS exists. Ship mode removes VSYS. 400 kHz, 3.31 V.
U1104
LMR38010SDDAR
L1103 XAL6060-223MEC 22uH
R1123 220k
R1124 43.2k
R1125 64.9k
C1161 4.7uF 100V
C1162 100nF 100V
C1163 100nF
C1164 22uF 25V
C1165 22uF 25V
C1166 22uF 25V
R1126 100k

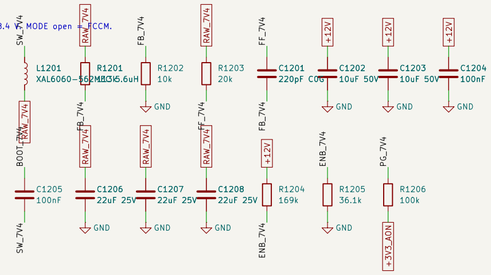
LM5050-1, LM5145 x2, LMR38010. Hand design, values in RATIONALE-log.
marengo
Sheet: /LP_logic_board/front_end_12v_24v/
File: front_end_12v_24v.kicad_sch
Title: front_end_12v_24v: ideal diode, hold-up, 12 V, 24 V, 3.3 V always-on
Size: A2 | Date: 2026-11-06 | Rev: A-wip
KiCad E.D.A. 10.0.8 | Id: 22/27

Servo rail. Adjust by R top: 90.9k = 6.05 V, 113k = 7.4 V, 130k = 8.4 V. MODE open = FCCM.

U1201

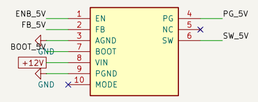


TPS56637RPAR

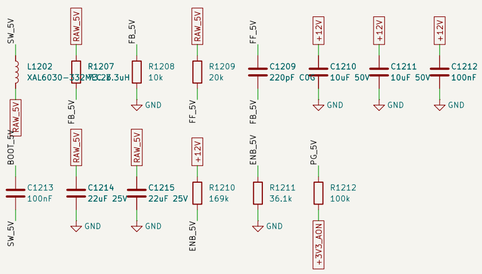


5 V peripheral, 4 A. EN divider per TI: starts near 10 V input.

U1202

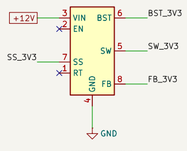


TPS56637RPAR

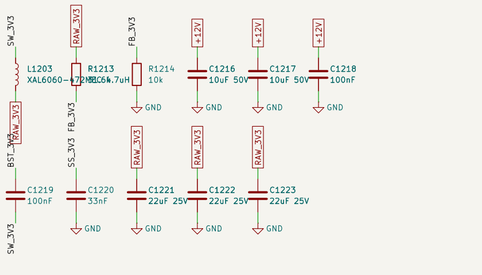


3.3 V peripheral, 2 A. RT open = 500 kHz, EN open = on.

U1203

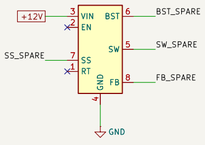


TPS62933DRLR

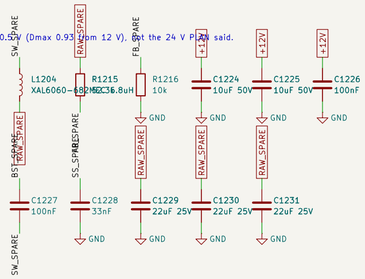


Spare rail, stuffed 5 V. 3.3 V: R top 31.6k. 9 V: see log. Max about 10.5 V (Dmax 0.93 from 12 V), but the 24 V PSU said.

U1204

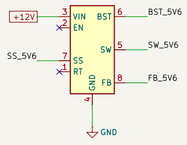


TPS62933DRLR

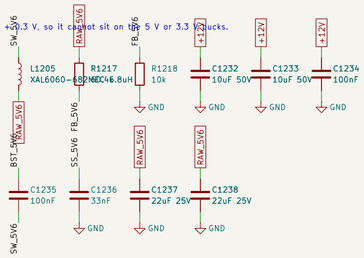


New: 5.6 V pre-regulator. TPS7A20 needs VIN <= 6.0 V and >= VOUT +0.3 V, so it cannot sit on the 5 V or 3.3 V bucks.

U1205

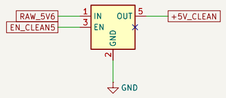


TPS62933FDRLR

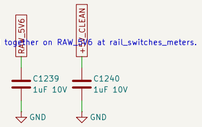


Clean rails, 0.3 A each, under 5 mV ripple. Metered together on RAW_5V6 at rail_switches_meters.

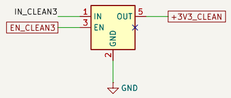
U1206



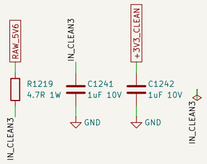
TPS7A2050PDBVR



U1207



TPS7A2033PDBVR





TPS56637 x2, TPS62933 x3, TPS7A20 x2. Hand design, values in RATIONALE-log.

marengo

Sheet: /LP_logic_board/pol_rails/

File: pol_rails.kicad_sch

Title: pol_rails: point-of-load rails from 12 V

Size: A2	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 23/27

OUT_24V: meter on bus A, straps A1,A0 = GND,GND, shunt 10mR

U1305 TPS26631PWPR — +12V; +24V; EN_24V; UV_24V; OV_24V; SW_24V; RAIL_FLT; IL_24V; DV_24V; GND

R1301 4.99k · R1302 475k · R1303 7.87k · R1304 22.1k · C1301 22nF · C1302 1uF 50V · C1303 100nF 50V · C1304 100nF 50V · R1305 33k

R1306 10mR 3W · R1307 10R · R1308 10R · C1305 100nF · C1306 100nF

U1306 INA238AIDGSR — +3V3_AON; OUT_24V; SP_24V; SN_24V; I2C_A_SDA; I2C_A_SCL; RAIL_ALERT_A; EN_12V_THIRD

OUT_12V_THIRD: meter on bus A, straps A1,A0 = GND,SDA, shunt 10mR

U1307 TPS26631PWPR — +12V; +24V; EN_12V_THIRD; UV_12V_THIRD; OV_12V_THIRD; SW_12V_THIRD; RAIL_FLT; IL_12V_THIRD; DV_12V_THIRD; GND

R1317 6.04k · R1318 442k · R1319 21.5k · R1320 41.2k · C1313 47nF · C1314 1uF 50V · C1315 100nF 50V · C1316 100nF 50V · R1321 33k

R1322 10mR 3W · R1323 10R · R1324 10R · C1317 100nF · C1318 100nF

U1308 INA238AIDGSR — +3V3_AON; OUT_12V_THIRD; SP_12V_THIRD; SN_12V_THIRD; I2C_A_SDA; I2C_A_SCL; RAIL_ALERT_A; EN_12V_PI

OUT_12V_PI_STUB: meter on bus A, straps A1,A0 = VS,GND, shunt 10mR

U1309 TPS26631PWPR — +12V; +24V; EN_24V; UV_12V_PI; OV_12V_PI; SW_12V_PI; RAIL_FLT; IL_12V_PI; DV_12V_PI; GND

R1333 4.99k · R1334 442k · R1335 21.5k · R1336 41.2k · C1325 47nF · C1326 1uF 50V · C1327 100nF 50V · C1328 100nF 50V · R1337 33k

R1338 10mR 3W · R1339 10R · R1340 10R · C1329 100nF · C1330 100nF

U1310 INA238AIDGSR — +3V3_AON; OUT_12V_PI_STUB; SP_12V_PI; SN_12V_PI; I2C_A_SDA; I2C_A_SCL; RAIL_ALERT_A; EN_5V_PERIPH

OUT_5V_PERIPH: meter on bus B, straps A1,A0 = GND,VS, shunt 5mR

U1313 TPS22965DSGR — RAW_5V; EN_5V; CT_5V; SW_5V; VBIAS; GND

C1337 1uF 10V · C1338 1nF · C1339 10uF 10V · R1349 100k

F1301 2.5A MF-MSMF250/16X · R1350 2.55k · R1351 10R · R1352 10R · C1340 100nF · C1341 100nF

U1311 INA238AIDGSR — +3V3_AON; OUT_5V_PERIPH; SP_5V; SN_5V; I2C_B_SDA; I2C_B_SCL; RAIL_ALERT_B; EN_SPARE

OUT_SPARE: meter on bus B, straps A1,A0 = GND,SCL, shunt 10mR

U1312 TPS26631PWPR — RAW_SPARE; SW_SPARE; UV_SPARE; OV_SPARE; EN_SPARE; RAIL_FLT; IL_SPARE; DV_SPARE; GND

R1357 4.99k · R1358 365k · R1359 33.2k · R1360 97.6k · C1347 47nF · C1348 1uF 50V · C1349 100nF 50V · C1350 100nF 50V · R1361 33k

R1362 10mR 3W · R1363 10R · R1364 10R · C1351 100nF · C1352 100nF

U1314 INA238AIDGSR — +3V3_AON; OUT_SPARE; SP_SPARE; SN_SPARE; I2C_B_SCL; I2C_B_SDA; RAIL_ALERT_B; EN_7V4

R1368 100k · R1369 10k · R1370 10k

TPS26631 UV/OV chain mandatory (UVLO to GND means 15.46 V, would kill a 12 V rail). Latch mode, supervisor clears. SHDN default off.
TPS22965 has no current limit. Its 6 A is under the buck limit; polyfuse after each switch. 5 V 2.5 A, 3.3 V 1.1 A.
Output limits: 24 V 3.61 A, Jetson 2.98 A, aux 3.97 A, PI stub 3.61 A, 7.4 V 5.98 A, spare 3.61 A (TPS26631 latch off), 5 V 2.5 A, 3.3 V 1.2 A (polyfuse, hold at 23 C, derates hot).
INA238 has no energy registers: firmware integrates at 100 Hz, or swap to INA228 (same footprint).

OUT_12V_JETSON: meter on bus A, straps A1,A0 = GND,VS, shunt 10mR

U1303 TPS26631PWPR — +12V; EN_12V_JETSON; UV_12V_JETSON; OV_12V_JETSON; SW_12V_JETSON; RAIL_FLT; IL_12V_JETSON; DV_12V_JETSON; GND

R1309 6.04k · R1310 442k · R1311 21.5k · R1312 41.2k · C1307 47nF · C1308 1uF 50V · C1309 100nF 50V · C1310 100nF 50V · R1313 33k

R1314 10mR 3W · R1315 10R · R1316 10R · C1311 100nF · C1312 100nF

U1304 INA238AIDGSR — +3V3_AON; OUT_12V_JETSON; SP_12V_JETSON; SN_12V_JETSON; I2C_A_SDA; I2C_A_SCL; RAIL_ALERT_A; EN_12V_AUX

OUT_12V_AUX: meter on bus A, straps A1,A0 = GND,SCL, shunt 10mR

U1302 TPS26631PWPR — +12V; EN_12V_AUX; UV_12V_AUX; OV_12V_AUX; SW_12V_AUX; RAIL_FLT; IL_12V_AUX; DV_12V_AUX; GND

R1325 4.53k · R1326 442k · R1327 21.5k · R1328 41.2k · C1319 47nF · C1320 1uF 50V · C1321 100nF 50V · C1322 100nF 50V · R1329 33k

R1330 10mR 3W · R1331 10R · R1332 10R · C1323 100nF · C1324 100nF

U1315 INA238AIDGSR — +3V3_AON; OUT_12V_AUX; SP_12V_AUX; SN_12V_AUX; I2C_A_SCL; I2C_A_SDA; RAIL_ALERT_A; EN_7V4_SERVO

OUT_7V4_SERVO: meter on bus B, straps A1,A0 = GND,GND, shunt 5mR

U1301 TPS26631PWPR — RAW_7V4; EN_7V4; UV_7V4; OV_7V4; SW_7V4; RAIL_FLT; IL_7V4; DV_7V4; GND

R1341 3.01k · R1342 383k · R1343 64.9k · R1344 64.9k · C1331 47nF · C1332 1uF 50V · C1333 100nF 50V · C1334 100nF 50V · R1345 33k

R1346 5mR 3W · R1347 10R · R1348 10R · C1335 100nF · C1336 100nF

U1316 INA238AIDGSR — +3V3_AON; OUT_7V4_SERVO; SP_7V4; SN_7V4; I2C_B_SDA; I2C_B_SCL; RAIL_ALERT_B; EN_5V_PERIPH

OUT_3V3_PERIPH: meter on bus B, straps A1,A0 = GND,SDA, shunt 10mR

U1315 TPS22965DSGR — RAW_3V3; EN_3V3; CT_3V3; SW_3V3; VBIAS; GND

C1342 1uF 10V · C1343 1nF · C1344 10uF 10V · R1353 100k

F1302 1.1A MF-MSMF110/16X · R1354 2.55k · R1355 10R · R1356 10R · C1345 100nF · C1346 100nF

U1317 INA238AIDGSR — +3V3_AON; OUT_3V3_PERIPH; SP_3V3; SN_3V3; I2C_B_SDA; I2C_B_SCL; RAIL_ALERT_B; EN_RAW_5V6_M

RAW_5V6_M: meter on bus B, straps A1,A0 = VS,GND, shunt 10mR

R1365 10mR 3W · R1366 10R · R1367 10R · C1353 100nF · C1354 100nF

U1318 INA238AIDGSR — +3V3_AON; RAW_5V6_M; SP_CLEAN; SN_CLEAN; I2C_B_SDA; I2C_B_SCL; RAIL_ALERT_B

TPS26631 eFuse / TPS22965 switch + INA238 per output. 2 I2C buses, 5 meters each.

marengo

Sheet: /LP_logic_board/rail_switches_meters/

File: rail_switches_meters.kicad_sch

Title: rail_switches_meters: rail switches and meters

Size: A1 | Date: 2026-11-06 | Rev: A-wip

KiCad E.D.A. 10.0.8 | Id: 24/27

E-stop loop, 6.9 mA: +12V, 680R, NC contacts, opto A, opto B, 620R, GND. Open loop, lost 12 V or lost 3V3 all give EN low.

J2001

J2002

J2003

tether, shorting dummy

wireless, shorting plug

E-stop loop

U2001

ISOM8710

U2003

ISOM8710

U2002

SN74LVC1G14DBV

U2004

SN74LVC1G14DBV

Q2001

2N7002

Q2002

2N7002

R2001 680R 0.25W

R2002 620R 0.25W

D2001 1N4148W

D2002 SMAJ15A

D2003 SMAJ15A

C2001 100nF

C2002 100nF

C2003 100nF

C2004 100nF

R2003 0R Cat 0 link

R2004 1k

R2005 100k

R2006 1k

R2007 160k

C2005 2.2uF

R2008 1M

D2004 BAT54

R2009 0R Cat 0 link

R2010 1k

R2011 100k

R2012 1k

R2013 160k

C2006 2.2uF

R2014 1M

D2005 BAT54

SUP_OPEN high: 2N7002 sinks EN. MCU dead or Hi-Z: pull-up opens the banks. The FET only sinks, so firmware cannot close an open loop.
Cat 1 option: remove the 0R, fit the four DNP parts. 289 ms nominal (232–356 ms). DNP in rev A: Cat 0.
Not Cat 3: one series loop feeds both channels, a jumping short across it defeats both.

Fans, 12 V, 25 kHz PWM open drain (polarity inverts in firmware).

J2004

J2005

J2006

fan 1

fan 2

fan 3

Q2003

2N7002

Q2004

2N7002

Q2005

2N7002

R2015 10k

R2016 10k

R2017 10k

R2018 100k

R2019 100k

R2020 100k

User I/O: 6-pin shell cable, the plan's 4 pins cannot carry RGB + button. Bench jumper, buzzer, charge LED, Qwiic, air NTC.

J2007

J2008

J2009

J2010

J2011

J2012

shell: button + RGB

charge LED

bench jumper

Qwiic

intake NTC

exhaust NTC

R2021 330R

R2022 330R

R2023 330R

R2024 330R

R2025 10k

R2026 10k

D2006 BAT54

R2027 100k

TP2001

TP2002

D2007 1N4148W

Q2006

2N7002

LS2001

PS1240P02BT

Rail outputs, one Micro-Fit each. TS2 wake: button low pulls AFE TS2 to VSS through 10k + BAT54 with the board off.

J2013

J2014

J2015

J2016

J2017

J2018

J2019

J2020

J2021

J2022

J2023

OUT_24V

OUT_12V_JETSON

OUT_12V_THIRD

OUT_12V_AUX

OUT_12V_PI_STUB

OUT_7V4_SERVO

OUT_5V_PERIPH

OUT_3V3_PERIPH

OUT_SPARE

5V_CLEAN

3V3_CLEAN

E-stop path: loop current to opto to inverter to gate-driver enable. No firmware in that path.

marengo

Sheet: /LP_logic_board/io_fans_estop_loop/

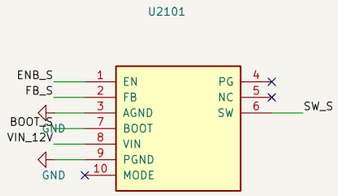
File: io_fans_estop_loop.kicad_sch

Title: io_fans_estop_loop: E-stop loop, fans, user I/O, output connectors

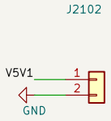
Size: A2	Date: 2026-11-06	Rev: A-wip
KiCad E.D.A. 10.0.8		Id: 25/27



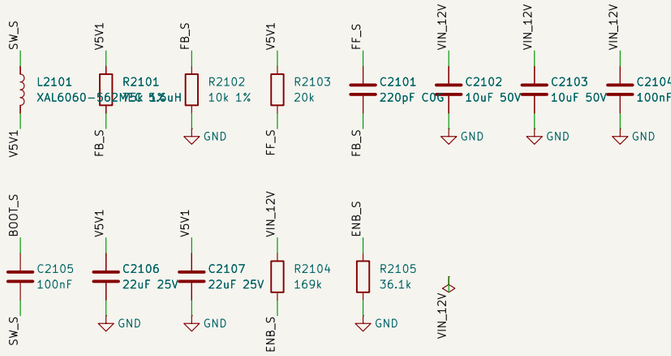
12 V in



TPS56637RPAR



5.1 V out, 8.5 A per contact



Output pigtail goes to a USB-C plug. A 2x3 housing on the GPIO header would put all 6 A through one GND pin (pin 6) and cannot lock.
Set usb_max_current_enable=1 on the Pi. PG left open: no logic rail on the stub.

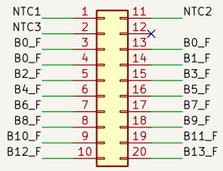
Own 2-layer board. Local nets; harness map checked in check.py.

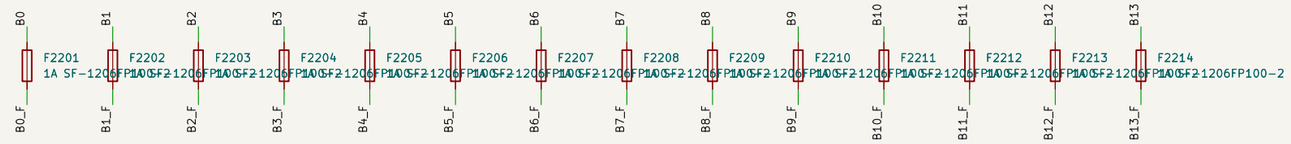
marengo

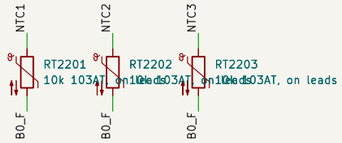
Sheet: /pi_power_stub/
File: pi_power_stub.kicad_sch

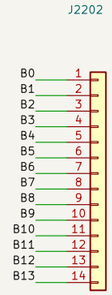
Title: pi_power_stub: 12 V to 5.1 V 6 A at the Pi

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 26/27









Fuse at the cell, one per tap. Presence: pin 13 tied to the B0 return. 1 A tap: 0.1455 Ohm, 63 V, 50 A interrupt (below the pack short-circuit current: rely on wire impedance or a higher breaking part, open).
Balance current through the fuse adds about 44 mV of sense error at 150 mA across two taps.

Own 2-layer board at the pack. Pin map must match HP J2. Local nets; harness map checked in che

marengo

Sheet: /pack_sense_board/
File: pack_sense_board.kicad_sch

Title: pack_sense_board: tap fuses, NTC, presence loop

Size: A3	Date: 2026-11-06	**Rev: A-wip**
KiCad E.D.A. 10.0.8		Id: 27/27